

Exploratory Data Analysis on the Titanic Dataset

Introduction

The Titanic dataset contains information about passengers aboard the Titanic, including demographic and personal details such as age, gender, ticket class, and survival status. This report presents an exploratory data analysis (EDA) to extract meaningful patterns, trends, and relationships within the dataset using Python and data visualization libraries such as Matplotlib and Seaborn.

Dataset Overview

- **Total entries:** 891
- **Total columns:** 12
- **Target variable:** Survived (1 = Survived, 0 = Did not survive)

Missing Values:

Column	Missing Values
Age	177
Cabin	687
Embarked	2

Univariate Analysis

a. Survival Count

- More passengers died than survived.
- Survived: 342
- Did not survive: 549

b. Gender Distribution

- Male: 577

- Female: 314
- A higher percentage of females survived compared to males.

c. Pclass Distribution

- 3rd class: 491
- 1st class: 216
- 2nd class: 184
- 1st class had the highest survival rate, while 3rd class had the lowest.

Bivariate Analysis

a. Survival by Gender

- Females had a significantly higher survival rate.
- Majority of males did not survive.

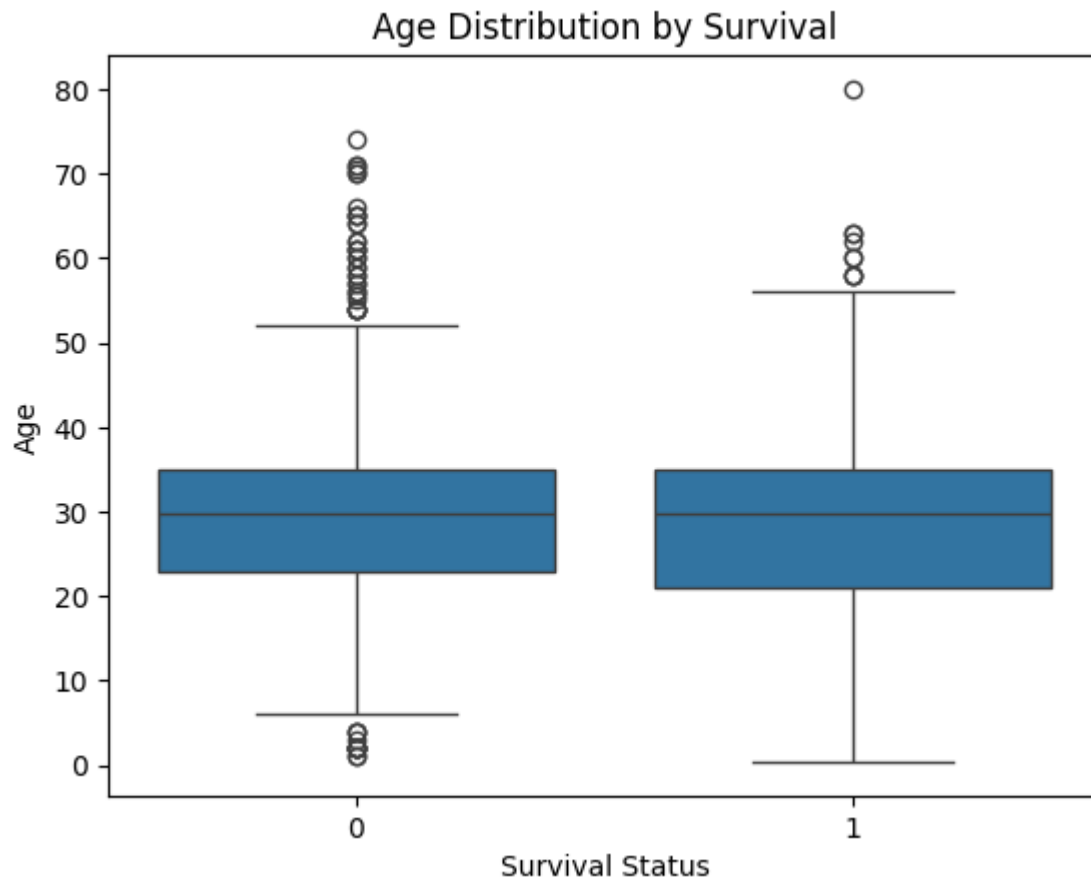
b. Survival by Pclass

- 1st class: High survival rate
- 3rd class: Very low survival rate

c. Survival by Age (Boxplot)

The boxplot below shows the distribution of passenger ages across survival status.

- Children, especially those under 10, had higher chances of survival.
- Many older passengers did not survive.
- Outliers are present in both survived and not-survived groups.



Multivariate Analysis

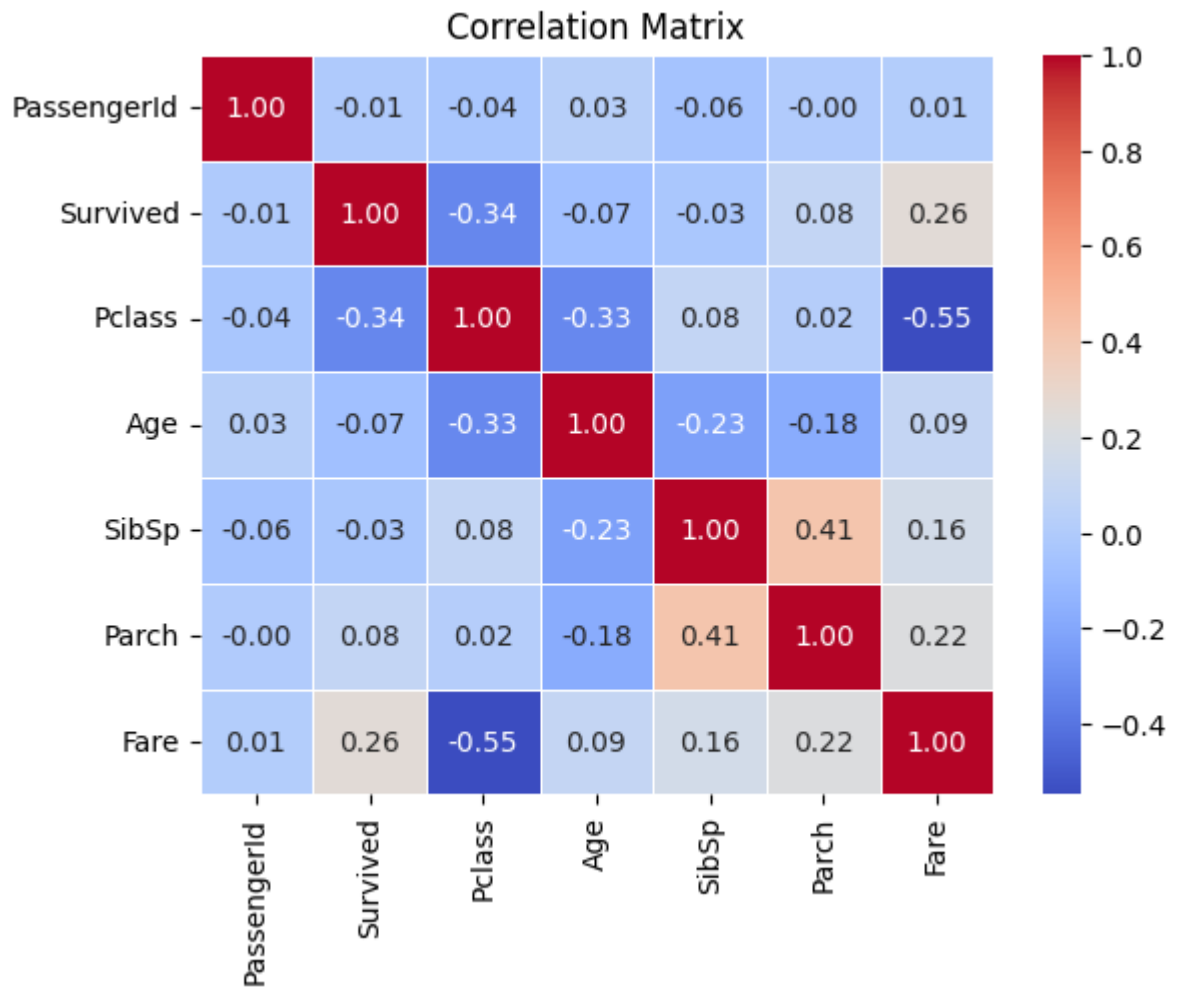
a. Correlation Heatmap

Feature	Correlation with Survived
Fare	+0.26
Pclass	-0.34
SibSp	-0.03
Parch	+0.03

- Positive correlation between **fare** and **survival**
- Negative correlation between **Pclass** and **survival** (lower class → lower chance)

b. Pairplot: Age, Fare, Survived

- Higher fares are mostly associated with survival.
- Survivors were often younger.



Key Insights and Patterns

- **Sex** and **Pclass** are strong predictors of survival.
- **Females** and **1st class** passengers were more likely to survive.
- **Children and younger passengers** had slightly better chances.
- **Higher ticket fares** were linked to higher survival.
- **Cabin data** is sparse and mostly missing.

Conclusion

This EDA provides a clear understanding of which factors impacted survival during the Titanic disaster. The most influential features include **Sex, Pclass, Age, and Fare**. Such insights are valuable for training machine learning models and highlight the importance of demographic features in predictive analysis.